

MUHAMMET EMİN AYHAN

AI & Robotics Engineer · Perception, Navigation & Robot Learning

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SUMMARY

Computer engineer (Selçuk University, 2026) building perception, navigation and learning systems for robots and UAVs: vision-language-action fine-tuning (SmolVLA / LeRobot), GPS-denied localization (neural dead reckoning, visual odometry, Kalman / EKF), real-time detection and ROS 2 autonomy on NVIDIA Jetson, measured on held-out data and shipped as Docker / CUDA services, ROS 2 nodes and REST APIs. **TEKNOFEST 2026 AI in Aviation finalist**; available immediately.

CORE SKILLS

Languages: Python, C++, Java, Dart, Swift, C#, SQL
Robotics & Autonomy: ROS 2 Humble, Nav2, SLAM (slam_toolbox), robot_localization, Gazebo, MuJoCo, Jetson Orin, STM32
Navigation & Estimation: Kalman / EKF, sensor fusion, visual odometry (RAFT optical flow, homography), neural dead reckoning
Machine Learning: PyTorch, VLA fine-tuning (SmolVLA, LoRA), imitation learning, LSTM / GRU / TCN, scikit-learn, XGBoost
Computer Vision: YOLO11 / YOLOv8, pose estimation, ByteTrack, EasyOCR / ALPR, SAHI, DINOv3, OpenCV
Engineering & Tools: Docker (CUDA), TorchScript, Flask REST, Git, Linux; Spring Boot, PostgreSQL, Flutter

EXPERIENCE

AI / ML Engineer Intern Jan 2026 – Jun 2026
ABBTech Bilgi Teknolojileri ve Danışmanlık — İstanbul, Türkiye
• Developed Kalman / EKF state estimation on IMU data for position and velocity in GPS-denied conditions; validated in simulation.
• Built Python data-processing and ML prototyping pipelines for decision-support systems.

Mobile App Developer Intern Jul 2025 – Aug 2025
Entegre Yazılım — Konya, Türkiye
• Cross-platform Flutter app with a Spring Boot REST backend (JWT, role-based authorization, PostgreSQL, Cloudinary media storage).

Backend Engineer Intern Jun 2025 – Jul 2025
Enoca — Konya, Türkiye
• Built Spring Boot REST APIs (Controller–Service–Repository, PostgreSQL / JPA) with Spring Security + Keycloak RBAC.

iOS App Developer Jul 2024 – Sep 2024
Presidency of the Republic of Türkiye, Digital Transformation Office — Ankara · Swift, SwiftUI, MVVM

KEY PROJECTS

Teaching Cost Curve — Robot Learning with Vision-Language-Action Models Independent research · 2026
• LoRA fine-tuned SmolVLA (450M, 0.66% of weights trained) on unseen LIBERO manipulation tasks on a single 8 GB GPU: success rose from **0% to 66.7% with only 5 demonstrations** (76.3% with 40; 300 episodes per point, Wilson 95% CIs).
• Showed that teaching a new task erases the old one (38% → 0%); fully reproducible pipeline guarded by invariant tests.

Aerial AI — GPS-Free Visual Odometry from UAV Imagery TEKNOFEST 2026 AI in Aviation · Finalist
• Owned the GPS-free positioning task (40% of the score): RAFT optical flow + homography with a keyframe ladder; **3.54 m mean error** over a full 2,256-frame rehearsal, 14× below the hold-position baseline.

GPS-Denied UAV Localization — Neural Dead Reckoning Independent project · 2026
• LSTM on 19 GPS-free sensor channels predicts per-second displacement: **44.4 m mean error after 4.5 min without GPS**, ~38× better than classical dead reckoning, ≈1 ms CPU inference (TorchScript).
• Found and removed a hidden GPS leak (GPS-fused yaw) in the inputs; barometer fusion cut vertical error by 45%.

Roadside Driver-Behaviour Analytics TEKNOFEST 2026 5G & Smart Road Safety
• YOLO11 detection & tracking, license-plate recognition (EasyOCR + voting) and pose-geometry driver-action detection from night-time windshield video; **precision 1.00 with zero false positives**; offline Docker / CUDA 12.1 service.

PATHIKA — Autonomous Ground Vehicle TEKNOFEST 2026 Unmanned Ground Vehicle
• ROS 2 Humble on Jetson Orin NX: owned SLAM, slalom planner, control and FSM mission manager; Nav2, EKF, STM32 e-stop.

MACHINOVA — ML-Based CFRP Surface-Roughness Prediction TUSAŞ Lift Up · Graduation project
• Five regressors on a Taguchi L9 design + ANOVA; RSM model **CV R² = 0.978** (RMSE 0.012 µm); desktop app, .exe and REST API.

EDUCATION & TRAINING

B.Sc. in Computer Engineering — Selçuk University, Konya 2022 – 2026 · GPA 3.56 / 4.00
Artificial Intelligence Specialization Program — National Technology Academy (Ministry of Industry and Technology) Jan 2026 – Aug 2026
Autonomous Driving Technologies Specialization — National Technology Academy Oct 2024 – Jun 2025

HONORS & COMPETITIONS

TEKNOFEST AI in Aviation	Finalist · 2026	BTK Academy Hackathon	Finalist · 2026
TEKNOFEST Air Defense Systems	Finalist · 2025	TEKNOFEST × Hepsiburada AI Hackathon	Finalist · 2025
INGHub Churn Prediction Datathon	Finalist · 2025	TÜBİTAK 2209-A Research Project	Funded · 2025
TEKNOFEST Flying Car Simulation	6th place · 2024		

CERTIFICATIONS & LANGUAGES

Certifications: Deep Learning for Image Processing & NLP (BTK Academy) · Swift Coding (Digital Transformation Office)
Languages: Turkish (Native) · English (B2)